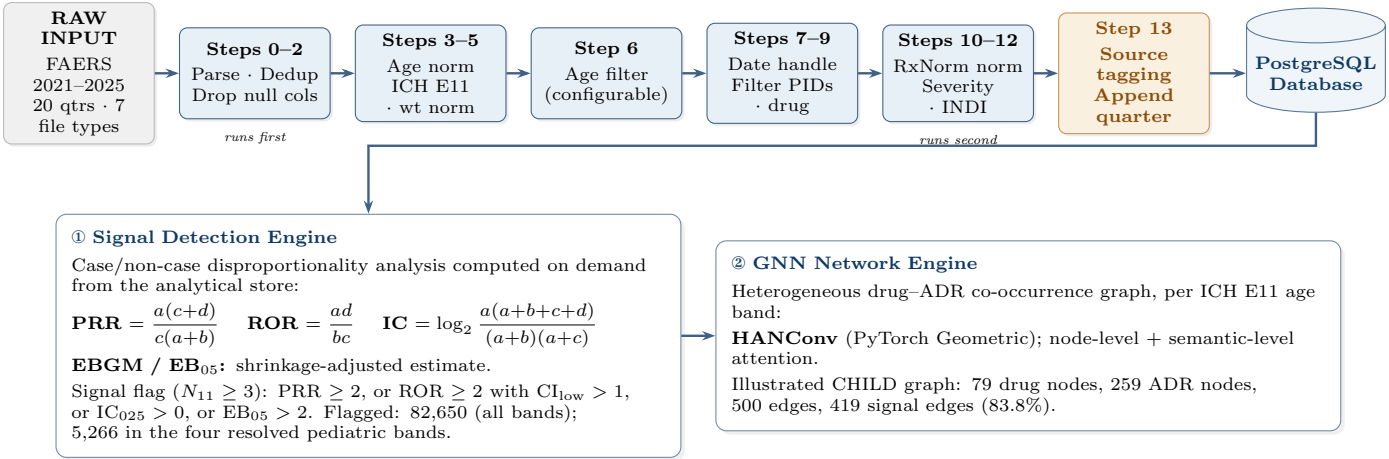

RAW INPUT
FAERS 2021–2025 20 qtrs · 7 file types
Steps 0–2
Parse · Dedup Drop null cols
runs first
Steps 3–5
Age norm ICH E11 · wt norm
Step 6
Age filter (configurable)
Steps 7–9
Date handle Filter PIDs · drug
Steps 10–12
RxNorm norm Severity · INDI
runs second
Step 13
Source tagging Append quarter
PostgreSQL Database
① Signal Detection Engine
Case/non-case disproportionality analysis computed on demand from the analytical store:
PRR = a(c+d)/c(a+b) ROR = ad/bc IC = log2 a(a+b+c+d)/(a+b)(a+c)
EBGM / EB05: shrinkage-adjusted estimate.
Signal flag (N11 ≥ 3): PRR ≥ 2, or ROR ≥ 2 with CIlow > 1, or IC025 > 0, or EB05 > 2. Flagged: 82,650 (all bands); 5,266 in the four resolved pediatric bands.
② GNN Network Engine
Heterogeneous drug–ADR co-occurrence graph, per ICH E11 age band:
HANConv (PyTorch Geometric); node-level + semantic-level attention.
Illustrated CHILD graph: 79 drug nodes, 259 ADR nodes, 500 edges, 419 signal edges (83.8%).